

Content

Learning objectives and content

Objectives

- The student is able to give examples of biomedical applications; can describe electronic systems used for these applications and can explain the function of the parts in such a system. The student can derive specifications for each building block based on the application goal.
- The student can follow a systematic design process (V-model), and can apply this in a design assignment. This process starts from application, requirements, system specifications, sub-block specifications, block implementation, up to verification and validation.
- The student can give examples of low-power design techniques on system, circuit and technology level; can explain their principles; and can apply these into a design.
- The student can provide examples of circuit blocks for the acquisition of neural signals and for the stimulation of neural tissues, and can illustrate the main design trade-offs characterizing circuits for neural interfaces.
- The student is aware of, can explain and can apply the techniques discussed during the lectures, including analog, mixed-signal and digital circuit blocks such as amplifiers, filters, and ADCs.
- The student can make basic layouts, can explain and apply layout techniques, and can explain the effect of parasitics.
- The student is able to use IC design tools for schematic and layout (Cadence or similar tool).

Content

This course starts with an introduction to biopotential and neural acquisition and stimulation. After describing the respective applications, the required electronic systems for these interfaces are explained (at high level), and practical requirements and challenges are illustrated. A summary of other applications in the healthcare domain where electronics play a prominent role is also given.

In the next part of the course, electronic system and circuit design is dealt with (bottom-up), towards building entire biopotential and neural interfaces. The course focuses on a systematic design approach (using the V-model), power-efficient design of key building blocks (including amplifiers, filters, digital circuitry and ADCs), and attention to particular circuit design challenges in the context of biopotential and neural interfaces.

After learning the system and circuit basics, the third part of the course is dedicated to applying this in the context of neural signal acquisition and neural tissue stimulation.

Throughout the course, a large design assignment is worked on, where a biopotential or neural interface is developed from system-level down to circuit-level, starting with application requirements, continuing with system optimization and circuit implementations, and finishing with system verification.

2026

Biopotential and neural interface circuits (5XCC0)

Practical information

Course module

5XCC0

Course type

Bachelor College

Credits (ECTS)

5 EC

Category

Advanced

Language of instruction

English

Offered by

Electrical Engineering - Integrated Circuits

Short promotional description of the course

This course gives an overview of low-power IC design techniques in the healthcare domain, with special focus to interfaces for the acquisition of neural signals and stimulation of the neural system. It also gives you the opportunity to practice these concepts in several circuit design assignments. Contents: Health applications, electronics fundamentals, low power design principles, low power analog and digital design, amplifiers, filters, ADCs, layout techniques, biomedical circuits, bio-inspired circuits, neural interfaces, Cadence software tutorials (schematic and layout), design assignments.

Course enrolment

OSIRIS Student

You can register yourself via OSIRIS Student. To do so, go to Enroll / Course.

Enrolment periods

- Block 3

Timeslot(s)

A - Mo 1-4, We 9-10, Th 5-8

Enrolment period

15 November 2026 until 3 January 2027 23:59

Unenrolment

until 1 February 2027 23:59

Entrance requirements

Assumed previous knowledge

5ECD0 Electronic circuits 1

5ECE0 Electronic circuits 2

Instructional modes

Instructional modes

- Lecture with notebook

Remark

- Lecture with notebook

Remark

Tests

Tests

- Assignment
Interim examination

Test weight
20

Minimum grade

Assessment
Design assignments
- Written examination
Final examination

Test weight
80

Minimum grade
5.0

Test enrolment

OSIRIS Student

You can register yourself via OSIRIS Student. To do so, go to Enroll / Test.

Assignment

- Block 3
Enrolment period
15 November 2026 until 21 March 2027 23:59

Written examination

- Block 3
Enrolment period
15 November 2026 until 21 March 2027 23:59

Unenrolment
until 21 March 2027
- Block 4
Enrolment period
15 November 2026 until 6 June 2027 23:59

Unenrolment
until 6 June 2027

Lecturer(s)

Co-lecturer

- prof.dr.ir. E. Cantatore

Contactperson for the course

- dr.ir. P.J.A. Harpe

Responsible lecturer

- dr.ir. P.J.A. Harpe

Subject matter expert

- dr.ir. P.J.A. Harpe

Materials

Recommended material

- Lecture slides
- Each student must bring a laptop with Matlab installed

Do you want to know more?

dr.ir. P.J.A. Harpe can tell you more about this course.

Send an e-mail to p.j.a.harpe@tue.nl